

SPEED BREAKER POWER GENERATOR

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Abstract--Necessity drives innovation. As the world rapidly increases in population and vehicle usage, as well as energy demands, we are seeing a significant surge in the need for sustainable energy sources. Energy is vital for the survival of all living beings, and the scarcity of traditional energy supplies has brought further attention to renewable alternatives. With traffic volume doubling or even tripling in recent years, we have the opportunity to harness this movement for energy generation. This study explores the concept of "power generation through speed breakers," inspired by various natural mechanisms. It's based on the simple fact that considerable energy is produced when vehicles pass over speed bumps. By capturing this energy, we can create a viable power source. The kinetic energy from moving vehicles can be transformed into mechanical energy via a rack and pinion system. This mechanical energy is then converted into electrical energy using a generator and stored in a battery for later use. Additionally, a spring system is incorporated to return the mechanism to its initial state. The energy collected during daylight hours can be beneficial at night, powering streetlights, traffic signals, and other applications. This paper discusses the design and implementation of power generation through rack and pinion systems at speed breakers.

Keywords--Energy, Rack and pinion, Battery, Power generation, Traffic, Speed Breakers.

I. INTRODUCTION

Energy sources can be broadly categorized into two main types: conventional and renewable. For centuries, conventional energy sources such as coal, oil, and natural gas have fueled industrialization and power generation. In contrast, renewable energy sources are clean, sustainable, and environmentally friendly. These include hydropower, wind energy, geothermal energy, solar power, and tidal energy. While energy comes in various forms—including mechanical, thermal, and electrical—electrical energy is often favored. In recent years, there has been a significant shift from conventional to renewable energy generation due to the numerous benefits associated with the latter. This transition is particularly vital for countries with limited public energy resources. For example, lighting expressways at night presents a challenge, as it requires considerable power supply. In India, the automotive industry is one of the largest globally, with an annual production of 23.37 million vehicles in the financial year 2014-15, reflecting a growth of 8.68% from the previous year. This sector contributes 7.1% to the country's gross domestic product (GDP). Two-wheelers dominate the Indian automotive market, holding an 81% share, driven by an expanding middle class and a young population. Moreover, the increasing interest from companies in exploring rural markets has further supported the sector's growth. The overall passenger vehicle segment accounts for

13% of the market share. This trend lends credibility to the concept of using speed bumps for energy generation. This paper explores the mechanisms behind this type of power generation and the potential energy output of such a system. Additionally, it introduces an innovative idea to install energy-collecting mediums before the speed bumps, aiming to harness more energy while effectively slowing down vehicles.

II. ANALYSIS OF THE PROPOSED SYSTEM:

In today's world, power has become an essential requirement for human life. The availability of electricity and its per capita consumption are indicators of a nation's standard of living in modern society. Energy plays a vital role in every sector of a country's economy.



Fig 1. Traffic in Major cities

Currently, around 400 million people lack access to electricity due to the grid not extending to their regions. An additional challenge is the inconsistent power supply, marked by frequency fluctuations that arise from imbalances in load generation. These fluctuations occur as consumer demand rises and falls. Maintaining a stable frequency is critical for the operation of alternating current (AC) systems. In India, the rated frequency is 50.0 Hz, but keeping it close to this value consistently has been a significant issue. Poor power quality has detrimental effects on equipment performance and can impact large-scale generation capacity. Unlike many developed nations, India also struggles with inadequate road infrastructure, characterized by numerous speed bumps.

However, there's potential to harness energy from these very speed breakers. By installing a "Power Generation Unit from Speed Breakers," we could capture a substantial amount of

energy. This energy could be utilized for lighting along the roads, allowing us to redirect power typically consumed by these lights to rural areas, thereby improving access to electricity where it is desperately needed.

III. APPARATUS REQUIRED

a) Spur Gears: Two pairs of Spur Gears are used As shown in Fig2. Their specifications are Mentioned below in Table 1

Table 1: Specifications of Spur Gear

Spur gears	2
Number of Teeth	35 and 18
Addendum diameter	74.22mm and 43.3mm
Module	2.0mm
Face width at base	4.16mm and 4.3mm
Pressure angle	20full depth teeth
Material	Hardened steel

b) Sprocket wheels: 2 regular and 1 freewheeling wheels are used. Material – mild steel and stainless steel, Addendum diameter – 78mm and 182mm, Number of teeth – 16 and 44, pitch – 12.7mm



c) Free wheel: material – mild steel, addendum diameter – 50mm, number of teeth – 18, freewheeling- one direction, pitch – 14.6mm

d) Fly wheel: To reduce the thrust force and thereby reducing the fluctuations in system

e) chain drive: pitch- 12.7mm, Roller diameter – 7.95mm, width – 7.85mm, breaking load- 13,800N.

f) spring: The rack needs to reverse its downward movement and move upwards. This is accomplished using springs positioned beneath the rack. The chosen springs are designed to ensure that during the downward movement, the majority of the force from the vehicle's weight is directed to the pinion, allowing the spring to compress easily. Once the weight on the rack is removed, the spring should return the rack to its original position. The specifications for the springs include a wire diameter of 3.36 mm, a spring index of 15.76, a mean coil diameter of 51.05 mm, and a total of 8 active coils.

g) rack: The idea is to transform the linear motion generated by a speed breaker into rotary motion to harness power. This is achieved through a rack and pinion mechanism, where the rack engages with a sprocket to facilitate movement. To enhance both the strength and cost-effectiveness of the model, we have replaced the traditional rack and pinion system with a chain that is securely attached to a wooden block. This chain then works in conjunction with a sprocket that interfaces with it, creating a more robust solution.

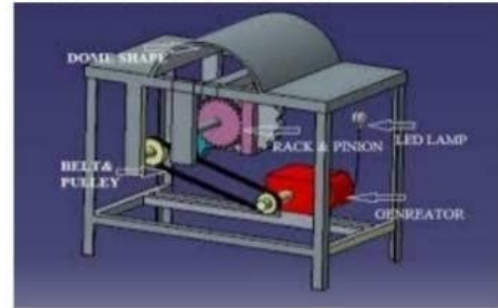


Fig 4. Setup of Rack and Pinion Mechanism using Belt and Pulley Arrangement

h) Shaft: An aluminum shaft with a diameter of 20mm serves as the mounting point for the gears. This shaft is supported by bearings, allowing it to rotate smoothly around its axis. The forces acting on the shaft include the weights of the components attached to it, the transmission loads, and the shaft's own weight.

i) Bearings: To facilitate the rotation of the shaft within the wooden frame, we've opted for single sleeve needle roller bearings, a specific kind of cylindrical roller bearing. The shaft's diameter has been reduced to match the inner diameter of the bearing, creating a step in front of the bearing. This design allows us to secure the shaft's movements along its axis and mitigates any potential eccentricity during rotation. Each set of bearings is responsible for supporting the weight of the components attached to the respective shaft, as well as handling the transmission loads.

IV. PRINCIPLE AND WORKING:

Electricity can be harnessed from speed breakers by implementing a gear system along with electronic components, leading to significant energy generation. The rack-pinion mechanism stands out for its ability to efficiently convert the energy generated into electricity through a straightforward design.

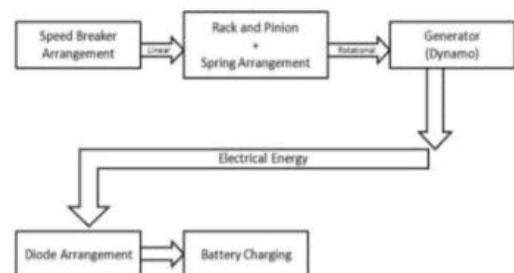


Fig 3. Flowchart of the Energy generation mechanism

In this setup, a load affects the speed breaker, which then transfers that load to the rack and pinion system. The motion created by the speed breaker is transformed into rotary motion via the rack and pinion mechanism. The axis of the pinion

rigid material, ensuring it won't impede the vehicle's weight. Fig 11 highlights the internal cavity designed for power-generating equipment and presents a front view of the spring mechanism. Additionally, a section cut is shown in Fig 11 to prevent the flap from dropping below the road surface.

Parts shown in Fig 10: flap, hinge, rack and pinion mechanism, spring housing, roller.

Parts shown in Fig 11: flap, roller, section to prevent flap from going below the road surface, spring, cavity for power generating components.

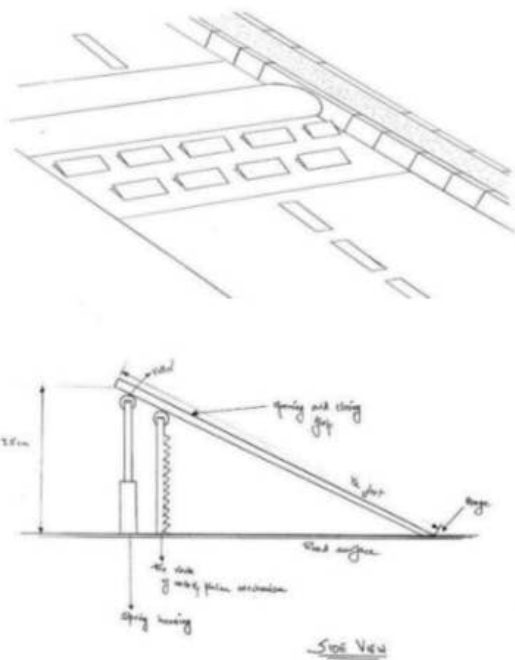


Fig 10. Side view of the Rack and Pinion Arrangement

VIII. POWER CALCULATION FOR THE NEW SYSTEM:

Consider,

The open end of flap to be at a distance of 7cm from the road surface

Length of the flap = 0.5 feet = 15.24 cm

Width of the flap = 1 feet = 30.48 cm

Mass of the vehicle = 1000kg

Calculations:

STEP 1: Force = Weight of the vehicle * acceleration due to gravity = $1000\text{Kg} \times 9.81 = 9810\text{N}$

STEP 2: Distance travelled by body = Height of speed brake = 7 cm

STEP 3: Output power = Work done/Sec = $(9810 \times 0.070)/60 = 11.445\text{W}$

STEP 4: Power developed for 1 vehicle passing over the 1 row of speed breaker arrangement in one minute is 11.445W

STEP 5: Power developed for 60 minutes (1 hour) = $11.445 \times 60 = 686.7\text{ W}$

STEP 6: Power developed for 24 hours = $686.7 \times 24 = 16480.8\text{W} = 16.4808\text{ KW}$

This values gets multiplied with the number of such rows of Rack and Pinion arrangement.

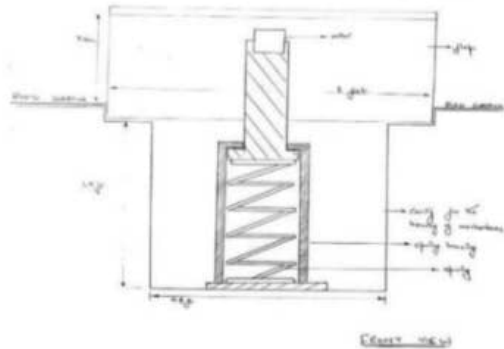


Fig 11. Front view of the Rack and Pinion Arrangement

ADVANTAGES OF RACK AND PINION ARRANGEMENT FOR POWER GENERATION:

Using this technology one can get the following benefits:

- Generation requires no manual intervention.
- Produces power without polluting the environment.
- Features straightforward construction and established technology.
- Maintenance is hassle-free.
- There are no issues related to fuel transportation.
- Energy is accessible throughout the entire year.
- Completely avoids the use of non-renewable fossil fuels.
- Maintenance costs are minimal.
- Installation expenses are low.
- Input frequency of vehicles remains consistently high.

APPLICATIONS:

The generated power is stored in the battery;

1. Street Lights: A street light, also known as a lamp-post or light standard, serves as a raised source of illumination along roads and walkways. These lights automatically turn on at a designated time each night. Most modern models are equipped with light-sensitive photocells, allowing them to switch on at dusk, turn off at dawn, or even activate during overcast conditions.

2. Traffic Lights: Commonly referred to as stoplights or signal lights, traffic lights are vital signaling devices found at intersections, pedestrian crossings, and other crucial points. Their primary function is to manage the flow of traffic by directing vehicles and pedestrians safely.

3. Calling Bells: This convenient mechanism can be installed at the entrance of homes. When someone steps onto

the doorstep, the weight triggers a signal, completing a circuit that activates the alert, notifying the occupants of a visitor.

IX. CONCLUSION:

With the increasing demand for electricity, new techniques are being developed to make effective use of available energy resources. One such innovative method is the rack and pinion system, which contributes to green power generation. This approach aims to enhance the world by utilizing resources more effectively. The best part? The system generates zero emissions, making it one of the eco-friendliest and reliable options available. The proposed mechanism is straightforward and continuously produces electrical power, offering a renewable energy source that is completely pollution-free. Additionally, this form of power generation is more cost-effective compared to many alternatives. Its design features fewer parts, leading to lower assembly costs, and all components can be sourced readily without the need for specialized manufacturing. Overall, this method represents a promising step toward sustainable energy solutions.

X. REFERENCES:

- [1] Aswathaman.V and Priyadharshini.M, every speed breaker is now a source of power, IPCBEE, VOL1, 2010.
- [2] Shakun Srivastava, Ankit Asthana, Department of Mechanical Engineering, Kanpur Institute of Technology,

Kanpur, Produce Electricity by the use of speed breakers,Journal of engineering research and studies.

- [3] Mohamad Ramadan, Mahmoud Khaled, Hicham El Hage, using speed bump for power generation—Experimental study, 7th International Conference on Applied Energy – ICAE2015.
- [4] Gurudath T V, Hemanth Raju Sundar, Bhagyalaxmi S Patil, Pavan Kumar R. – Kaunas University of Technology, Kaunas, Lithuania – International Research Journal of Engineering and Technology on 09 September 2019.
- [5] Sarada A, Department of Mechanical Engineering and Manufacturing Technology, Electricity Generation from Speed Breakers on (2023).
- [6] Mirsha et al, Rao et al,Speed Breaker Electricity Generator on (2013,2014).
- [7] S.Jaswanth Reddy,R.Pavankalyan,S.Waseem Akram, and V.keshava,Power Generation by using speed Breaker”at the international journal of emerging technologies and innovative research in June 2020.
- [8] Rajat Gupta,Suyash Sharma,and Saurabh Gaykawad,”A Revolutionary Technique of Power generation through speed Breaker power generators” at the international journal of engineering research and technology in 2013.